

Nonlinear Tucker Decomposition with Polynomial Links: Rank Inflation and Tensor Recovery

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Abstract—Many multidimensional signals admit compact low-rank tensor representations, but nonlinear measurements can make a low-rank signal appear to have a higher Tucker rank after observation. We first introduce a Tucker polynomial model, which extends the linear map from a Tucker core to the data tensor with higher-order interactions. Its general form learns new mode maps for each polynomial order. Sharing the same Tucker factors across orders yields NTD-PL, a compact form that is exactly a low-rank Tucker tensor followed by an entrywise polynomial response. The same estimator also supports other finite response bases. We analyze the resulting rank inflation, parameter cost, approximation of continuous responses, stationarity under standard descent conditions, and computational scaling. We also introduce a low-cost diagnostic that tests how much of a Tucker residual can be removed by refitting only the response. Experiments on controlled data, CAVE hyperspectral reconstruction and completion, noisy measurements, and several non-hyperspectral datasets show consistent improvements over matched-rank Tucker and competitive low-rank tensor baselines. Larger gains occur when the learned response increases the effective rank of the fitted tensor and when a response-only refit removes more of the Tucker residual.

Index Terms—Tensor decomposition, Tucker decomposition, low-rank tensor recovery, nonlinear measurements, tensor completion, hyperspectral imaging.

I. INTRODUCTION

MULTIDIMENSIONAL signals often contain strong structure across space, spectrum, time, view, or acquisition mode. Low-rank tensor models exploit this structure by representing a signal through a small number of coupled low-dimensional factors. They have become standard tools for signal representation, denoising, completion, and recovery [1]–[4]. Tucker decomposition is especially useful in this setting because it separates a compact core from mode-wise factors and provides an explicit multilinear rank budget.

In many sensing pipelines, however, the structured signal and the recorded data are not on the same scale. Camera response curves, radiometric processing, and saturation are direct examples of nonlinear transformations before data storage [5], [6]. Post-nonlinear hyperspectral mixing provides a related precedent in which a lower-dimensional signal is transformed after a structured mixing stage [7]–[9]. These processes have different physical origins, but they motivate the same question: what happens when a compact tensor signal is observed after a nonlinear response? Related questions also arise in generalized low-rank and single-index matrix models [10], [11].

For a fixed-rank Tucker model, this distinction matters because an entrywise nonlinear response can increase the

apparent multilinear rank of the measured tensor. A low-rank signal before the response may therefore require a much larger Tucker rank if the measured tensor is modeled directly. This creates a specific mismatch: the selected rank can be suitable for the latent signal while a Tucker model with the same rank is insufficient for the measurements.

Increasing the Tucker rank can reduce this mismatch, but it changes the whole multilinear representation: the core grows rapidly and every mode factor gains additional degrees of freedom. More general functional, kernel, and neural tensor models can describe a broader range of nonlinear effects [12]–[14]. Our goal is a more structured model that keeps a clear Tucker rank budget while allowing higher-order interactions.

We begin from the Tucker model itself. For fixed mode factors, standard Tucker maps the core linearly to the data tensor. We extend this map by introducing a *p-fold Tucker term*, which combines p copies of the core through higher-order mode maps. The case $p = 1$ is standard Tucker, and a sum over several orders gives a *Tucker polynomial*. This is a direct nonlinear extension of the core map, but the general form needs new mode maps at every order and quickly becomes large.

The model becomes compact when the same Tucker factors are shared across all orders. Let

$$\mathcal{S} = [\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}] \quad (1)$$

be the Tucker tensor formed by the shared factors. Under this sharing, the degree- p Tucker term is exactly $\mathcal{S}^{\odot p}$. The resulting model is

$$\hat{\mathcal{Y}} = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p} = f_{\beta}(\mathcal{S}), \quad (2)$$

which we call nonlinear Tucker decomposition with polynomial links (NTD-PL). Compared with the general Tucker polynomial, NTD-PL reuses one core and one set of factors and adds only the scalar coefficients $\beta = (\beta_0, \dots, \beta_P)$.

This derivation gives NTD-PL two complementary interpretations. It is the shared-factor form of the Tucker polynomial, and it is also a low-rank Tucker tensor followed by an entrywise polynomial response. The first view explains where the higher-order interactions come from. The second connects the model to nonlinear measurements. The same estimator can replace the powers by other finite sets of scalar basis functions when a different response basis is appropriate.

The Tucker parameters and response coefficients are estimated jointly from the observed entries of the measured tensor $\mathcal{Y}$. The objective fits available measurements and predicts missing entries while using $\mathcal{S}$ as a compact latent representation. Without external calibration constraints, the decom-

position into $\mathcal{S}$ and f_β need not identify a unique physical pre-response signal; the fitted pair is interpreted through its prediction of $\mathcal{Y}$.

Sharing the factors greatly reduces the parameter count, but it does not remove the high-order effect. Each $\mathcal{S}^{\odot p}$ has a Tucker representation whose factors contain degree- p monomials of the shared factors. The measured tensor can therefore have higher effective rank while the latent Tucker ranks remain fixed. The model is useful when these shared higher-order terms explain a substantial part of the Tucker residual. If they do not, increasing the Tucker rank or changing the tensor prior is a more direct remedy.

We fit the shared-factor model with a masked least-squares objective. The estimator alternates between a small ridge regression for the response coefficients and gradient-based updates of the Tucker core and factors. We also introduce a low-cost diagnostic: after fitting a linear Tucker model, we keep it fixed, refit only the response coefficients, and measure the residual reduction. A large reduction indicates that joint NTD-PL training is more likely to be useful.

A. Contributions

The main contributions are summarized as follows.

- We introduce the Tucker polynomial, which extends the linear core map of Tucker decomposition with p -fold interactions. Sharing the mode factors across orders yields NTD-PL, a compact model for nonlinear tensor measurements.
- We develop a masked alternating estimator for the shared-factor model, with a closed-form ridge update for the response coefficients and descent updates for the Tucker parameters. We also propose a diagnostic that measures how much of a fixed Tucker residual can be removed by refitting only the response.
- We establish the relation between p -fold Tucker terms and NTD-PL, analyze the induced rank inflation and parameter cost, prove approximation of continuous responses, and study stationarity and computational scaling.
- We evaluate controlled nonlinear measurements, CAVE reconstruction and completion, noisy reconstruction, and cross-domain diagnostics. Comparisons with matched-rank and higher-rank Tucker, CP, and TT test accuracy and parameter use, while mechanism experiments connect the gains to the measured rank increase and to the response diagnostic score.

Section II reviews related tensor and nonlinear observation models. Section III introduces the Tucker polynomial and derives NTD-PL, Section IV presents the estimator and diagnostic, Section V gives the analysis, and Section VI reports the experiments. Section VII concludes the paper.

II. RELATED WORK

A. Low-Rank Tensor Recovery

CP and Tucker decompositions are widely used for compression, denoising, source separation, completion, and data fusion [1], [2], [15]. Tucker decomposition represents a tensor

by a core and mode-wise factors [16], so its multilinear ranks directly specify the capacity assigned to each mode. Tensor-train and tensor-ring decompositions control complexity through different network structures [17], [18]. These formats provide useful alternative low-rank priors, while Tucker is the focus here because its mode-wise ranks let us compare the rank of the latent tensor with the rank observed after a nonlinear response.

Tensor completion methods recover missing entries by constraining the tensor with convex surrogates or factorized low-rank models. Representative approaches include nuclear-norm-based formulations, Tucker or CP factorizations, and nonconvex algorithms for incomplete tensors [3], [4], [19]. These methods generally place the low-rank constraint directly on the measured tensor. In our setting, the Tucker constraint is placed on a latent pre-response tensor, while the measured tensor can have a larger apparent rank. Increasing the Tucker rank remains an important baseline, but it adds parameters throughout the core and factors. NTD-PL instead tests whether the additional measured structure can be explained by a shared scalar response at the prescribed latent Tucker rank.

B. Polynomial and Nonlinear Tensor Models

Polynomial predictors can be written with high-order coefficient tensors. Tensor factorizations have been used to compress these coefficients and avoid forming all polynomial features explicitly [20], [21]. These methods produce scalar predictions from vector inputs. The Tucker polynomial considered here has a different role: it maps a Tucker core to a tensor output, while the mode maps preserve the structure of each output dimension. NTD-PL is the shared-factor form of this model.

Generalized tensor factorizations combine low-rank structure with a prescribed observation distribution or link. Generalized CP models support a range of loss functions and data types, while probabilistic Tucker models use application-specific likelihoods for observations such as counts [22], [23]. Nonparametric Bayesian tensor models provide another form of flexibility through latent-function priors [24]. In these approaches, the observation model is typically chosen as part of the statistical formulation.

Kernel, functional, and neural tensor models learn broader nonlinear feature maps or decoders [12]–[14], [25]. They are designed to capture nonlinear interactions that need not reduce to a single entrywise response. NTD-PL addresses a narrower case: one scalar function is shared across entries, and the input to that function is a Tucker tensor with explicitly prescribed ranks. This restriction retains a direct connection between the latent rank budget and the complexity added in the measured domain.

The closest matrix analogue is low-rank estimation with a generalized loss or an unknown entrywise transfer function. Generalized low-rank models cover many data-dependent losses, and monotonic single-index matrix completion jointly estimates a low-rank matrix and an unknown monotone link [10], [11]. The present work follows the same view for Tucker tensors: low-rank structure is modeled before an entrywise response. The Tucker polynomial supplies the higher-order

tensor map, while sharing its factors yields a small response model that can be fitted and diagnosed efficiently.

C. Nonlinear Responses in Imaging

Imaging pipelines provide established examples of nonlinear measurement responses. Camera response functions map scene irradiance to recorded pixel values, and high-dynamic-range imaging often estimates or calibrates this map explicitly [5], [6]. These models motivate learning a low-dimensional response instead of absorbing all radiometric variation into the image representation.

Hyperspectral imaging contains other forms of nonlinearity, especially in spectral mixing. Reviews of nonlinear unmixing and post-nonlinear mixing models describe settings in which a structured linear mixture is followed by a nonlinear transformation [7]–[9]. These physical mechanisms are not all equivalent to one global entrywise response. Their relevance here is the broader modeling precedent: a compact structured signal can be considered before a nonlinear observation stage.

Accordingly, NTD-PL is not presented as a complete physical model of a camera or hyperspectral sensor, and the real-data experiments do not claim to identify a unique calibration curve. Instead, we test whether a jointly learned shared response improves reconstruction and completion of the measured tensor at a fixed latent Tucker rank. Hyperspectral cubes are useful for this test because of their strong spatial and spectral correlations. The response diagnostic further measures, before joint fitting, whether a scalar response can reduce the residual of a Tucker model on a given tensor.

III. TUCKER POLYNOMIAL MODELS

A. Tucker as a Linear Core Map

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ denote the measured tensor and let $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$ be the observation mask. A Tucker tensor is

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad (3)$$

where $\mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}$ is the core and $\mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}$ are mode factors. Entrywise,

$$S_{\mathbf{i}} = \sum_{r_1, \dots, r_N} X_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (4)$$

For fixed factors, this is a linear map from the core to the data tensor. The core controls interactions among the mode coordinates, while the factors map those coordinates to the observed dimensions.

B. Tucker Polynomial Model

We extend the linear core map by allowing polynomial interactions among core entries. Throughout, P denotes the maximum polynomial degree and p indexes a degree. For $p \geq 1$, define a p -fold Tucker term by

$$[\mathcal{T}_p(\mathcal{X})]_{\mathbf{i}} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \prod_{\ell=1}^p X_{\mathbf{r}^{(\ell)}} \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (5)$$

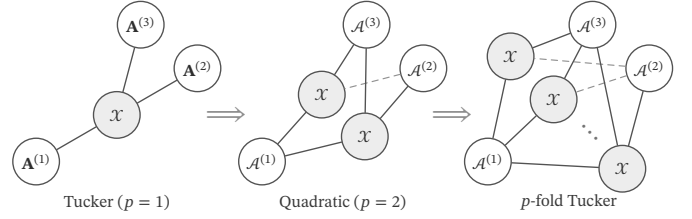


Fig. 1. Tensor-network view of the p -fold Tucker term in (5) for $N = 3$. Starting from Tucker, the quadratic and general p -fold terms connect two or p copies of the core to the same three mode maps. Dashed edges and the ellipsis reduce visual clutter.

Here $\ell = 1, \dots, p$ indexes the core copies and $\mathcal{A}^{(n,p)}$ is the order- p mode map for mode n . When $p = 1$, choosing $\mathcal{A}^{(n,1)} = \mathbf{A}^{(n)}$ recovers standard Tucker. For $p \geq 2$, the term is a degree- p polynomial in the core entries. Figure 1 visualizes these contractions for $N = 3$. The degree label on $\mathcal{A}^{(n,p)}$ is omitted inside the nodes to keep the diagram readable.

Summing several orders gives the Tucker polynomial

$$\hat{\mathcal{Y}} = \beta_0 \mathbf{1} + \sum_{p=1}^P \beta_p \mathcal{T}_p(\mathcal{X}). \quad (6)$$

The general model assigns a separate mode map to every order. It provides a direct polynomial extension of Tucker, but its parameter count grows quickly with p . It is the general starting model; the next subsection derives the compact form used for estimation.

C. Shared Factors and NTD-PL

We share the original Tucker factors across polynomial orders by setting

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{\ell=1}^p A_{i_n, r_n^{(\ell)}}^{(n)}. \quad (7)$$

Substituting this relation into (5) gives

$$\mathcal{T}_p(\mathcal{X}) = \mathcal{S}^{\odot p}. \quad (8)$$

The shared-factor Tucker polynomial is therefore

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad (9)$$

which is NTD-PL. The explicit expansion and its ordinary Tucker representation are given in the supplementary material.

Equation (9) also gives the nonlinear measurement interpretation

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad \mathbf{i} = (i_1, \dots, i_N), \quad (10)$$

where $\varepsilon_{\mathbf{i}}$ denotes measurement noise. Thus NTD-PL is both a compact Tucker polynomial and a low-rank Tucker tensor followed by a shared entrywise response. Sharing the factors removes the separate high-order mode maps; compared with Tucker, the only additional trainable variables are the $P + 1$ response coefficients.

The data determine the combined prediction $f_{\beta}(\mathcal{S})$. Without external calibration constraints, we do not claim that the fitted $\mathcal{S}$ and response recover a unique physical measurement process.

D. Other Response Bases

The estimator is not limited to powers. A finite set of scalar basis functions gives

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{j=0}^J \beta_j \psi_j(\mathcal{S}), \quad (11)$$

where $J + 1$ is the number of basis functions and $\psi_j(\mathcal{S})$ applies ψ_j entrywise. Standard Tucker is recovered when the fitted response is the identity. Possible alternatives include conditioned polynomial bases, radial basis functions, and splines. These choices retain the same Tucker representation and estimation objective. The p -fold Tucker derivation applies to the power basis, which is the main model used in this paper.

E. Masked Recovery Objective

The same formulation covers reconstruction and completion. We estimate the core, factors, and response coefficients by minimizing

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \mathcal{R}(\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta), \quad (12)$$

where $\mathcal{R}$ contains regularization terms for the Tucker parameters and the response coefficients. For reconstruction, Ω contains all entries. For completion, Ω selects the entries used for fitting, and the remaining entries are used for evaluation.

IV. ESTIMATION ALGORITHM AND RESPONSE DIAGNOSTIC

We estimate the shared-factor model in (11). The general Tucker polynomial in (6) motivates the construction, but its separate high-order mode maps are not part of the estimator.

Let $m = |\Omega|$ and let $\theta = (\mathcal{X}, \{\mathbf{A}^{(n)}\}_{n=1}^N, \beta)$ collect all trainable parameters. For an active response basis with $J + 1$ functions, we minimize

$$\mathcal{F}_J(\theta) = \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2, \quad (13)$$

where $f_{\beta}(s) = \sum_{j=0}^J \beta_j \psi_j(s)$ and $\{\psi_j\}_{j=0}^J$ is the selected response basis. For the power basis, $J = P$ and its p th function is $\psi_p(s) = s^p$, which gives (9); in implementation, the same polynomial space may be represented by centered, Chebyshev, or otherwise conditioned bases. Non-polynomial choices such as RBF or spline bases use the same objective as long as the response remains linear in β . The experiments use the monomial power basis. The estimation alternates between an exact small response-coefficient update and a descent step for the Tucker core and factors.

A. Response Refresh as a Ridge Subproblem

Fix the current Tucker signal $\mathcal{S}$ and list the observed entries as $y_{\Omega} \in \mathbb{R}^m$ and $s_{\Omega} \in \mathbb{R}^m$. Define the feature matrix $\Phi_{\Omega, J}(\mathcal{S}) \in \mathbb{R}^{m \times (J+1)}$ by

$$[\Phi_{\Omega, J}(\mathcal{S})]_{t, j} = \psi_j([s_{\Omega}]_t). \quad (14)$$

Here $t = 1, \dots, m$ indexes the observed entries and $j = 0, \dots, J$ indexes the basis functions. The response update is the ridge regression

$$\beta^+ = \arg \min_{\beta \in \mathbb{R}^{J+1}} \frac{1}{2m} \|y_{\Omega} - \Phi_{\Omega, J}(\mathcal{S})\beta\|_2^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2, \quad (15)$$

with closed-form solution

$$\beta^+ = \left(\frac{1}{m} \Phi_{\Omega, J}^T \Phi_{\Omega, J} + \lambda_{\beta} \mathbf{I} \right)^{-1} \frac{1}{m} \Phi_{\Omega, J}^T y_{\Omega}. \quad (16)$$

This step solves only a $(J+1) \times (J+1)$ system. It is therefore cheap relative to reconstructing or differentiating through the Tucker tensor, and because it solves the β -subproblem exactly, it never increases $\mathcal{F}_J$ for the current Tucker signal.

B. Descent for the Tucker Parameters

With β fixed, the residual is backpropagated through the entrywise response and then through the Tucker map. The gradient with respect to the Tucker signal is

$$\mathbf{G}_{\mathcal{S}} = \frac{1}{m} \Omega \odot (f_{\beta}(\mathcal{S}) - \mathcal{Y}) \odot f'_{\beta}(\mathcal{S}). \quad (17)$$

The Tucker gradients are standard reverse contractions:

$$\nabla_{\mathcal{X}} \mathcal{F}_J = \mathbf{G}_{\mathcal{S}} \times_1 (\mathbf{A}^{(1)})^T \cdots \times_N (\mathbf{A}^{(N)})^T + \lambda_X \mathcal{X}, \quad (18)$$

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{F}_J = [\mathbf{G}_{\mathcal{S}}]_{(n)} \mathbf{Z}_n^T + \lambda_n \mathbf{A}^{(n)}, \quad (19)$$

where $\mathbf{Z}_n = (\mathbf{A}^{(N)} \otimes \cdots \otimes \mathbf{A}^{(n+1)} \otimes \mathbf{A}^{(n-1)} \otimes \cdots \otimes \mathbf{A}^{(1)}) \mathcal{X}_{(n)}^T$. For the convergence result in Section V, the Tucker update is taken to satisfy a sufficient decrease condition, as with a line-search or suitable block-coordinate step. The experiments use Adam as a practical first-order solver.

C. Response-Basis Continuation and Normalization

Large or high-order response bases can amplify Tucker scale ambiguity. For nested bases, such as the power or Chebyshev bases, we use continuation: training starts with the Tucker-compatible identity response and activates additional basis functions gradually. When a new function is introduced, its coefficient is initialized at zero, so the prediction is unchanged at the moment of activation. Non-nested bases such as RBF and spline bases are trained with all selected basis functions from the start. After Tucker updates, factor columns are rescaled and the inverse scales are absorbed into the core. This fixes a numerical gauge without changing the represented signal $\mathcal{S}$ or the data-fitting term. It does not make the Tucker factors identifiable in a strict statistical sense; it only keeps the Tucker parameterization and response coefficients on stable numerical scales during training.

Algorithm 1 treats the response as part of the fitted representation: the Tucker update is computed through the current nonlinear measurement map, rather than by post-processing a fixed Tucker fit.

Algorithm 1 Alternating recovery for NTD-PL and other finite response bases.

Require: measured tensor $\mathcal{Y}$, mask Ω , Tucker ranks $(R_1, \dots, R_N)$, response basis $\{\psi_j\}$, regularization weights, continuation schedule

- 1: Initialize $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$ from a Tucker fit or a random Tucker factorization.
- 2: Set the active basis to the identity response and initialize $\beta = (0, 1, 0, \dots, 0)$.
- 3: **for** outer iterations $k = 0, 1, \dots$ **do**
- 4: Form the current Tucker signal $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$.
- 5: Build $\Phi_{\Omega, J}(\mathcal{S})$ on the observed entries.
- 6: Refresh β by the ridge solution in (16).
- 7: Compute $\mathbf{G}_{\mathcal{S}}$ from (17).
- 8: Update $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$ by a descent step using (17)–(19).
- 9: Normalize factor columns and absorb inverse scales into $\mathcal{X}$.
- 10: **if** the continuation schedule activates a new basis function **then**
- 11: Add the new response coefficient and initialize it at zero.
- 12: **end if**
- 13: **if** the stopping criterion is met **then**
- 14: Stop.
- 15: **end if**
- 16: **end for**

Ensure: fitted Tucker signal $\mathcal{S}$, response coefficients β , and prediction $f_{\beta}(\mathcal{S})$

D. Response Diagnostic

The diagnostic measures how much the residual of a linear Tucker fit decreases when only the entrywise response is refit. Let $\mathcal{S}_T$ be a fitted Tucker reconstruction, and let $\beta_T = (0, 1, 0, \dots, 0)$ denote the identity response in the degree- J basis. We first compute

$$\hat{\beta}_{\text{resp}} = \arg \min_{\beta} \|y_{\Omega} - \Phi_{\Omega, J}(\mathcal{S}_T)\beta\|_2^2 + m\lambda_{\beta} \|\beta\|_2^2. \quad (20)$$

The residual energies before and after this response-only fit are

$$R_T = \|y_{\Omega} - \Phi_{\Omega, J}(\mathcal{S}_T)\beta_T\|_2^2, \quad (21)$$

$$R_{\text{resp}} = \|y_{\Omega} - \Phi_{\Omega, J}(\mathcal{S}_T)\hat{\beta}_{\text{resp}}\|_2^2. \quad (22)$$

We report their ratio on a decibel scale:

$$D_{\text{link}} = 10 \log_{10} \left(\frac{R_T}{R_{\text{resp}}} \right) \text{ dB}. \quad (23)$$

A large positive D_{link} means that refitting only the response coefficients substantially reduces the residual without changing the Tucker fit. We use this score as a model-selection and interpretation tool: a small or nonpositive score suggests that increasing multilinear rank or changing the tensor prior may be more appropriate.

V. ANALYSIS

This section analyzes the shared-factor Tucker polynomial and its estimator. It first connects the general p -fold model to NTD-PL, then studies the resulting rank and parameter cost. The remaining results cover response approximation, descent, and computational scaling. The optimization results also apply to other differentiable finite response bases whose response is linear in the coefficients.

A. Tucker Polynomial Structure and Rank Inflation

We begin with the effect of sharing the mode factors across polynomial orders.

Proposition 1 (Shared-factor p -fold Tucker term). *Let $\mathcal{S}$ be the Tucker tensor in (3). If the mode maps of the degree- p term satisfy (7), then*

$$\mathcal{T}_p(\mathcal{X}) = \mathcal{S}^{\odot p}. \quad (24)$$

Consequently, the shared-factor Tucker polynomial is exactly NTD-PL.

This result explains why the two views of the model agree. The general p -fold term is a polynomial map of the core. Sharing its mode factors turns that term into an entrywise power of one Tucker tensor.

Proposition 2 (Response-induced Tucker rank inflation). *Let $\mathcal{S}$ have Tucker rank at most $(R_1, \dots, R_N)$. For every $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (25)$$

The quantity $\binom{R_n + p - 1}{p}$ is the number of degree- p monomials in a row of the mode- n factor. Thus each shared p -fold term has an ordinary Tucker representation with polynomially expanded factors, even though those factors are determined by the original ones.

Corollary 1 (Parameter cost of sharing). *The direct degree- p term in (5) contains $\sum_{n=1}^N I_n R_n^p$ mode-map entries. Let $D_n(p) = \binom{R_n + p - 1}{p}$. An ordinary Tucker model that explicitly represents the shared degree- p term may require*

$$d_{\text{lift}}(p) = \prod_{n=1}^N D_n(p) + \sum_{n=1}^N I_n D_n(p) \quad (26)$$

parameters for that term alone. The shared-factor model represents all degrees $0, \dots, P$ with the original Tucker parameters plus $P + 1$ response coefficients.

Sharing therefore reduces both constructions: it avoids separate p -fold mode maps and avoids storing the larger Tucker representation induced by the polynomial terms. This economy applies only to interactions generated from the same core and factors; the general Tucker polynomial remains more flexible.

Theorem 1 (Separation from matched-rank Tucker). *Let $N \geq 3$, $R \geq 2$, and $p \geq 2$. Define $D = \binom{R + p - 1}{p}$ and suppose $I_n \geq D$ for every mode. Then there exists an order- N tensor representable by a degree- p NTD-PL model with Tucker rank $(R, \dots, R)$, but whose exact standard Tucker rank is $(D, \dots, D)$.*

The theorem shows that sharing does not reduce the model to matched-rank Tucker. A shared p -fold term can attain the expanded ranks in Proposition 2, while the trainable Tucker tensor remains at rank $(R, \dots, R)$. This is an existence result, not a claim that NTD-PL spans all tensors of the larger rank.

B. Response Generation and Approximation

The previous results explain how a polynomial response can inflate the Tucker rank observed after measurement. The next two results ask what kinds of entrywise responses the power basis can represent on top of the same low-rank Tucker signal.

Proposition 3 (Exact generation for polynomial responses). *If $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P , then a degree- P NTD-PL model with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 2 (Continuous response approximation). *Let $\mathcal{S}^*$ have Tucker rank at most $(R_1, \dots, R_N)$ and entries in a compact interval $[a, b]$. If the measurement response f is continuous on $[a, b]$, then for every $\epsilon > 0$ there exists a polynomial degree P and coefficients β such that*

$$\|f(\mathcal{S}^*) - f_\beta(\mathcal{S}^*)\|_F \leq \epsilon\sqrt{M}, \quad M = \prod_{n=1}^N I_n. \quad (27)$$

The exact polynomial result is a useful special case. The approximation result is broader: the power response is not restricted to exactly polynomial sensors. On a bounded signal range, it can approximate any smooth or continuous entrywise response to the desired uniform accuracy, while keeping the Tucker rank of the underlying signal fixed. The degree required for a given accuracy depends on the regularity of the response and the width of the signal range. This approximation result is the reason the power basis is used as the default response basis in the experiments.

C. Descent and Stationarity

We analyze the fixed-basis objective $\mathcal{F}_J$ in (13). For nested bases, the continuation schedule can be viewed as a sequence of such fixed-basis problems, each initialized from the previous basis size.

Assumption 1. *For a fixed active basis size $J + 1$, the iterates remain in a bounded normalized parameter set. The regularization weights are nonnegative and $\lambda_\beta > 0$. The Tucker step either uses a line search or an equivalent descent rule such that, for constants $c > 0$ and $b > 0$,*

$$\mathcal{F}_J(\theta^k) - \mathcal{F}_J(\theta^{k+1}) \geq c \|\theta^{k+1} - \theta^k\|_2^2, \quad (28)$$

$$\text{dist}(0, \partial \mathcal{F}_J(\theta^{k+1})) \leq b \|\theta^{k+1} - \theta^k\|_2. \quad (29)$$

The boundedness condition is enforced in practice by Tucker gauge normalization and regularization. The two descent conditions are standard for block coordinate and line-search gradient schemes on smooth objectives.

Proposition 4 (Monotone objective decrease). *Under Assumption 1, every fixed-basis outer iteration of the alternating estimator satisfies*

$$\mathcal{F}_J(\theta^{k+1}) \leq \mathcal{F}_J(\theta^k). \quad (30)$$

Moreover, if (28) holds for the entire outer update, then

$$\sum_{k=0}^{\infty} \|\theta^{k+1} - \theta^k\|_2^2 < \infty. \quad (31)$$

Theorem 3 (Convergence to a stationary point). *Suppose Assumption 1 holds for a fixed basis size $J + 1$. Then the sequence generated by the alternating estimator has finite length and converges to a critical point θ^* of $\mathcal{F}_J$:*

$$0 \in \partial \mathcal{F}_J(\theta^*). \quad (32)$$

The theorem does not claim global optimality; the problem remains nonconvex because of the Tucker factorization. It applies to the descent-controlled implementation specified in Assumption 1. The Adam solver used in the experiments is a practical implementation and is not claimed to satisfy these conditions at every iteration. Under the stated conditions, the result places the alternating scheme on the same theoretical footing as standard nonconvex block-descent methods [26], [27].

D. Computational Scaling

Let $M = \prod_{n=1}^N I_n$, $R_\Pi = \prod_{n=1}^N R_n$, and $d_{\text{Tucker}} = R_\Pi + \sum_{n=1}^N I_n R_n$. Storing the fitted model requires

$$d_{\text{NTD}} = d_{\text{Tucker}} + (J + 1) \quad (33)$$

parameters. Thus the learned response adds only $J + 1$ parameters to the Tucker model. For one outer iteration, Response Refresh forms $\Phi_{\Omega, J}^T \Phi_{\Omega, J}$ and solves a small system, costing

$$\mathcal{O}(mJ^2 + J^3). \quad (34)$$

The dominant cost remains the Tucker reconstruction and reverse contractions; evaluating the response adds $\mathcal{O}(MJ)$ work for dense tensors or $\mathcal{O}(mJ)$ on observed entries. Detailed scaling comparisons, the projection interpretation of D_{link} , and a masked-entry generalization bound are provided in the supplementary material.

VI. EXPERIMENTS

A. Experimental Questions and Protocol

The experiments are organized around three questions. First, can the linked model recover tensors whose observations contain a nonlinear response? Second, under the same Tucker rank, does this response improve real hyperspectral reconstruction and completion? Third, when the model helps, do the gains follow the mechanism predicted by the analysis, and can this regime be detected before fitting NTD-PL?

Tucker is the primary baseline because a comparison at the same Tucker rank directly isolates the effect of the learned scalar response. We also compare with CP and tensor train (TT) using similar parameter counts. These methods provide broader low-rank tensor baselines and show whether the improvement extends beyond one Tucker implementation.

For reconstruction, RMSE and SAM are computed on all tensor entries. For completion, RMSE* and SAM* are computed only on entries hidden during fitting. Hyperspectral cubes are globally normalized, and lower RMSE and SAM are better. The primary real dataset is the CAVE multispectral image database [28], [29]. All CAVE experiments use all 15 scenes, each of size $512 \times 512 \times 31$; Fig. 2 shows their RGB previews. We also include supplementary checks on external

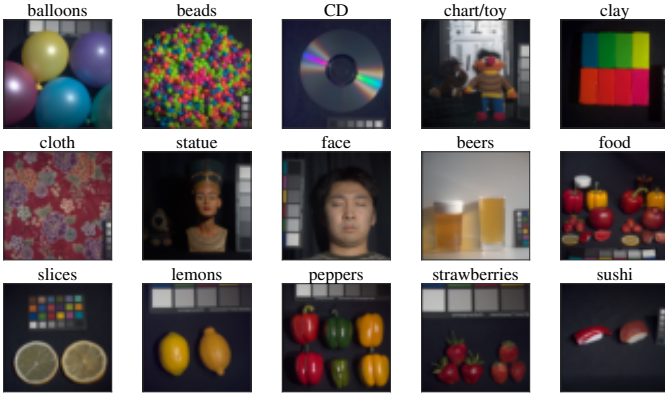


Fig. 2. CAVe scenes used in the hyperspectral recovery experiments. Each tile shows the database-provided RGB preview of a 512×512 multispectral cube.

hyperspectral cubes and on non-hyperspectral tensor data, including natural images, object-view tensors, and action videos, to test whether the same response-based pattern appears beyond the CAVe acquisition setting. Additional robustness, dataset, and protocol details are provided in the supplementary material.

B. Controlled Recovery Under Nonlinear Measurements

The controlled experiment tests the basic mechanism in a setting where the ground-truth Tucker structure is known. We start from an underlying Tucker signal $\mathcal{S}^*$ of known rank and observe every tensor entry. For nonlinear responses, we add a response component that is orthogonal to the underlying Tucker signal. Let $\tilde{\mathcal{S}}$ denote the normalized underlying signal and let $\tilde{\mathcal{R}}$ denote the normalized nonlinear residual after this projection. The noisy measurement is generated as

$$\mathcal{Y}_\alpha = \sqrt{1 - \alpha} \tilde{\mathcal{S}} + \sqrt{\alpha} \tilde{\mathcal{R}} + \mathcal{E}, \quad (35)$$

where α controls the energy assigned to the nonlinear residual, and $\mathcal{E}$ is additive Gaussian noise. The identity panel directly uses $\tilde{\mathcal{S}}$ for every α , so it is a repeated noisy linear control. This construction keeps the intended Tucker rank fixed while making the measured tensor increasingly mismatched to a linear Tucker observation model.

Figure 3 shows the expected transition. We use 10 random seeds, Tucker rank $(4, 4, 4)$, and additive Gaussian noise at 30 dB SNR for all panels. In the identity-control panel, Tucker and NTD-PL are nearly identical, with mean RMSE about 0.029 for both methods across the sweep. Thus the linked response does not create a spurious advantage when the measurement is already linear.

The behavior changes once a nonlinear residual is present. As α increases, matched-rank Tucker must fit a measured tensor whose apparent rank has grown, while NTD-PL can reuse the same Tucker signal through a learned scalar response. The gap is therefore small in the nearly linear regime and widens for stronger nonlinear measurements; for example, in the $\tanh(\kappa s)$ panel at $\alpha = 0.4$, Tucker reaches RMSE 0.338 whereas NTD-PL remains near 0.124. This pattern is the desired mechanism check: the advantage appears when

the observation departs from a linear Tucker measurement map, not because the model has extra coefficients. Degree-dependent controls and optimization traces are reported in the supplementary material.

C. Real Hyperspectral Recovery on CAVe

We next test whether NTD-PL improves real spectral-spatial tensors.

Reconstruction first tests whether the learned response improves accuracy at a fixed Tucker rank. We compare NTD-PL with Tucker, CP, and TT at similar parameter counts. Table I compares the methods at approximately 13k, 27k, and 42k parameters. The CP and TT ranks are adjusted at each budget, and NTD-PL is reported for $P = 2, 4, 6$. Increasing P consistently lowers both RMSE and NMSE at every budget, with a larger improvement from $P = 2$ to $P = 4$ and a smaller refinement from $P = 4$ to $P = 6$. Thus the same degree-saturation pattern holds across model capacities rather than only at one selected rank.

NTD-PL with $P = 6$ gives the lowest mean RMSE at all three budgets. Relative to the strongest alternative at each budget, its RMSE reduction is 11.1% at 13k and 6.1% at 27k; at 42k it is essentially tied with CP and remains below Tucker. The NMSE results have the same ordering at the low and medium budgets, whereas CP has the best high-budget NMSE. RMSE and NMSE are averaged scene-wise, but NMSE first normalizes each scene by its own energy, so their order need not coincide. The narrowing gap is expected: as the base tensor rank grows, ordinary low-rank models can absorb more of the measured structure and the marginal value of the response decreases.

Across the tested Tucker ranks, NTD-PL also consistently improves matched-rank Tucker in both RMSE and SAM, with larger gains at lower ranks and smaller gains as the Tucker rank grows. This is the expected behavior of a fixed-rank nonlinear correction: if ordinary multilinear rank is already high enough to absorb the nonlinear response, the marginal value of the response should decrease.

Table II asks a complementary capacity question: how much ordinary Tucker rank is needed to match a lower-rank NTD-PL model? Across the tested ranks, Tucker needs roughly 51–60% more parameters to match the NTD-PL RMSE, while NTD-PL still has lower SAM. The result supports the rank-inflation interpretation in Section V: the power response reuses the same low-rank Tucker signal to express structured nonlinear residuals with a small number of response coefficients.

Completion then tests whether this fixed-rank benefit transfers to entries not used for fitting. Table III reports CAVe completion under random masks using all 15 scenes. At the same Tucker rank, NTD-PL improves RMSE* by 12.5–17.7% and SAM* by 13.4–29.3% across the tested missing rates. Additional observation-noise results are reported in the supplementary material.

D. Mechanism: Rank Inflation and Learned Response

The Tucker polynomial view says that higher degrees add interactions among the same core coordinates, while the

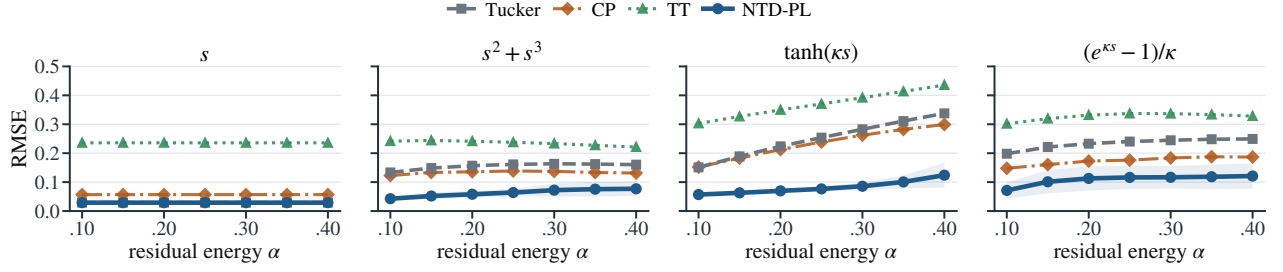


Fig. 3. Controlled noisy recovery under identity and nonlinear responses. All panels use additive Gaussian noise at 30 dB SNR and 10 random seeds. For the nonlinear panels, α controls the energy of the component orthogonal to the underlying Tucker signal; the identity panel s serves as a repeated linear control.

shared-factor form expresses those terms as powers of one Tucker tensor. We examine this mechanism by varying the maximum degree P on the CAVE scenes while keeping the Tucker rank at $(12, 12, 4)$. This lets us track the rank added by the higher-order terms and the learned scalar map $s \mapsto f_{\beta}(s)$.

For each scene and each $P \in \{2, 3, 4, 5, 6\}$, we fit a matched-rank Tucker model and an NTD-PL model using all entries of the normalized cube. Let $\hat{S}$ be the Tucker signal reconstructed from the NTD-PL core and factors, and let $\hat{Y}$ be its fitted prediction after the scalar response. We isolate the nonlinear response component $\hat{R}_{nl} = \hat{Y} - \hat{\beta}_0 \mathbf{1} - \hat{\beta}_1 \hat{S}$ and compute singular-value spectra of its mode-1 and mode-2 unfoldings. The rank reported in Fig. 4 is the number of singular values needed to capture 99.9% of squared singular-value energy; it is therefore an energy-based effective rank, rather than the exact algebraic rank. The bars show the mean RMSE reduction of NTD-PL relative to matched-rank Tucker across the 15 scenes, while the annotations show the median nonlinear-response rank and the median effective-rank increase over $\hat{S}$, respectively.

Figure 4 shows the rank-inflation trend. At $P = 2$, the nonlinear component already has nonzero effective rank, but its energy is not yet sufficient to increase the median rank of the final prediction. A median rank increase of 2.0 appears at $P = 3$, then increases gradually to 5.0 at $P = 6$, while the mean RMSE gain rises from 2.82% to 8.79%. The largest change occurs between $P = 2$ and $P = 4$; higher degrees provide smaller refinements, while the learned Tucker signal remains at the prescribed rank. Across scenes and degrees, spatial tail capture by the NTD-PL prediction has Spearman correlation 0.86 with RMSE gain, indicating that the improvement is tied to how much high-rank residual

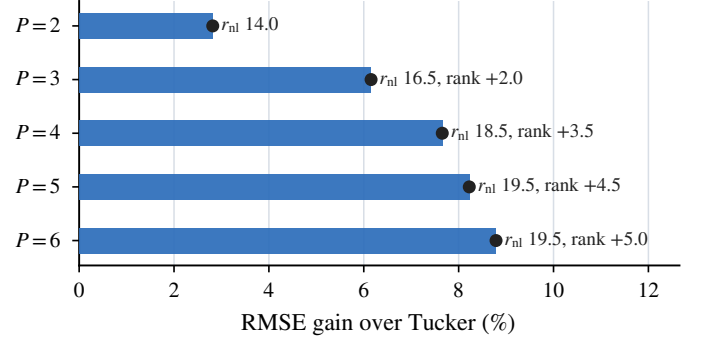


Fig. 4. Rank-inflation evidence on CAVE reconstruction at Tucker rank $(12, 12, 4)$. Bars show the mean RMSE reduction relative to matched-rank Tucker. Labels report r_{nl} , the median 99.9%-energy rank of the nonlinear response component, and, when positive, the median effective-rank increase over the NTD-PL Tucker signal.

structure the response accounts for.

Figure 5 shows the corresponding learned response curves. For each scene and each P , we evaluate the fitted polynomial at 101 empirical quantiles between the 1st and 99th percentiles of the learned Tucker signal. Each curve is then normalized by its endpoint values, so both axes run from 0 to 1 and the dashed diagonal is the identity response. Thin curves show individual scenes and the dark curve is their pointwise mean. The title gain is the mean RMSE reduction relative to the $P = 1$ model. The $P = 1$ model is linear by construction. Allowing $P = 2$ and $P = 4$ bends the shared response away from the identity map and yields most of the gain; $P = 6$ mainly refines the same shape. Detailed mechanism tables are given in the supplementary material.

TABLE I
CAVE RECONSTRUCTION ACROSS MATCHED PARAMETER BUDGETS AND RESPONSE DEGREES. ENTRIES ARE MEAN $\pm$ STANDARD DEVIATION OVER 15 SCENES.

Method	Low ($\approx 13k$)		Medium ($\approx 27k$)		High ($\approx 42k$)	
	RMSE $\downarrow$	NMSE(dB) $\downarrow$	RMSE $\downarrow$	NMSE(dB) $\downarrow$	RMSE $\downarrow$	NMSE(dB) $\downarrow$
CP	0.0333 $\pm$ 0.0271	-15.73 $\pm$ 5.05	0.0233 $\pm$ 0.0223	-19.44 $\pm$ 5.79	0.0186 $\pm$ 0.0191	-21.69 $\pm$ 6.11
Tensor Train	0.0437 $\pm$ 0.0292	-12.98 $\pm$ 4.70	0.0376 $\pm$ 0.0279	-14.53 $\pm$ 4.98	0.0334 $\pm$ 0.0266	-15.72 $\pm$ 5.18
Tucker	0.0330 $\pm$ 0.0265	-15.76 $\pm$ 4.97	0.0248 $\pm$ 0.0219	-18.58 $\pm$ 5.41	0.0208 $\pm$ 0.0184	-20.16 $\pm$ 5.52
NTD-PL ($P = 2$)	0.0308 $\pm$ 0.0251	-16.39 $\pm$ 4.78	0.0232 $\pm$ 0.0197	-18.98 $\pm$ 5.05	0.0197 $\pm$ 0.0160	-20.29 $\pm$ 4.98
NTD-PL ($P = 4$)	0.0296 $\pm$ 0.0250	-16.86 $\pm$ 4.99	0.0221 $\pm$ 0.0195	-19.52 $\pm$ 5.19	0.0189 $\pm$ 0.0157	-20.73 $\pm$ 5.05
NTD-PL ($P = 6$)	0.0293 $\pm$ 0.0249	-16.96 $\pm$ 4.98	0.0219 $\pm$ 0.0193	-19.63 $\pm$ 5.17	0.0186 $\pm$ 0.0156	-20.85 $\pm$ 5.03

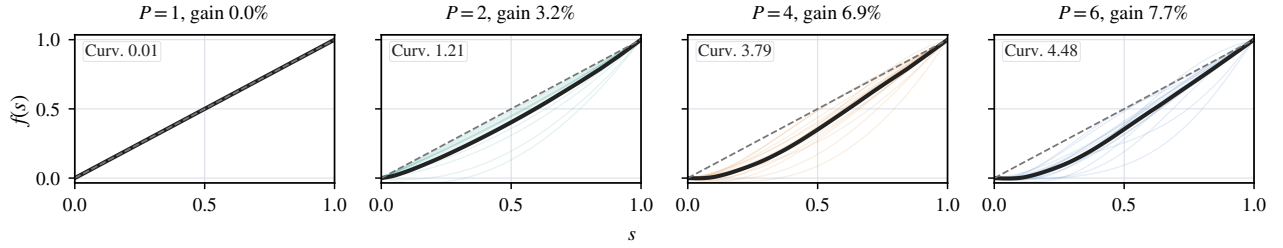


Fig. 5. Learned scalar response curves on CAFE reconstruction. Curves are evaluated on the 1st–99th percentile range of the learned Tucker signal and normalized by their endpoint values. Thin curves are scene-level fits and the dark curve is their pointwise mean. The title reports mean RMSE reduction relative to $P = 1$; “Curv.” is the normalized mean absolute curvature averaged across scenes.

TABLE II
TUCKER RANK NEEDED TO MATCH NTD-PL RMSE ON CAFE RECONSTRUCTION. NTD-PL USES $P = 6$. THE PARAMETER COLUMN SHOWS NTD-PL $\rightarrow$ TUCKER; SAM IS REPORTED AS NTD-PL/TUCKER.

NTD-PL	Tucker	Params	SAM
(18, 18, 3)	(28, 28, 3)	20k $\rightarrow$ 31k (+59.5%)	19.23 /21.41
(24, 24, 4)	(35, 35, 4)	27k $\rightarrow$ 41k (+51.3%)	15.15 /16.26
(33, 33, 4)	(49, 49, 4)	38k $\rightarrow$ 60k (+56.5%)	14.41 /15.28

TABLE III
CAVE COMPLETION UNDER RANDOM MASKS AT TUCKER RANK (33, 33, 4) AND $P = 4$. RMSE* AND SAM* ARE AVERAGED OVER THE ENTRIES HIDDEN DURING FITTING.

Missing	RMSE $\downarrow$			SAM $\downarrow$		
	Tucker	NTD-PL	Rel. gain	Tucker	NTD-PL	Rel. gain
0.1	0.0263	0.0229	12.9%	11.82	8.36	29.3%
0.3	0.0263	0.0230	12.5%	14.75	12.47	15.5%
0.5	0.0264	0.0230	12.9%	15.56	13.48	13.4%
0.7	0.0266	0.0219	17.7%	16.02	13.79	13.9%

E. Operating Regimes and Diagnostics

The previous experiments explain why NTD-PL helps when the Tucker residual is response-like. We finally ask whether this condition can be detected before joint NTD-PL training, and where the gain appears when the model is useful.

On CAFE reconstruction, the error reduction is not uniform over the cube. When entries are grouped by normalized observed value and spectral regime, the largest gains occur in middle intensity deciles and the 510–700 nm regimes, while a few low-yield bins remain near zero. This supports the same operating-regime view: the response is useful when the Tucker residual has coherent structure in the observed domain rather than unstructured noise. Figure 6 visualizes the marginal and joint regimes; the supplementary material reports the corresponding numerical table.

The diagnostic D_{link} is computed by fitting a Tucker model and measuring, in dB, the residual-energy reduction obtained by refitting only the response coefficients. It is therefore substantially cheaper than joint NTD-PL training and is used here as a model-selection signal.

On CAFE reconstruction, treating each scene as one tensor unit gives a similar trend: high- D_{link} scenes have larger eventual NTD-PL gains than low- D_{link} scenes. The CAFE row is reported together with the cross-domain diagnostic statistics in the supplementary material.

Figure 7 evaluates the same diagnostic beyond hyperspectral data. Each point is an independently fitted tensor unit, and the horizontal axis is D_{link} measured before joint NTD-PL training. Object-view tensors show the clearest separation, action-video tensors follow the same trend, and natural images occupy a lower-yield but still positive regime. These results close the loop: NTD-PL is most effective when a matched-rank Tucker fit leaves residual structure that can be reduced by a shared entrywise response.

VII. CONCLUSION

This paper introduced the Tucker polynomial, which extends the linear map from a Tucker core to the data tensor with p -fold interactions. Sharing the mode factors across polynomial orders yields NTD-PL. This compact form reuses one core and one set of factors, and it is also a low-rank Tucker tensor followed by an entrywise polynomial response. Other finite scalar response bases can be used with the same estimator when different response bases are appropriate.

The analysis established the relation between the general p -fold terms and their shared-factor form, quantified the resulting rank and parameter cost, and studied response approximation, nonconvex descent, and computational scaling. The estimator combines a small ridge update for the response coefficients

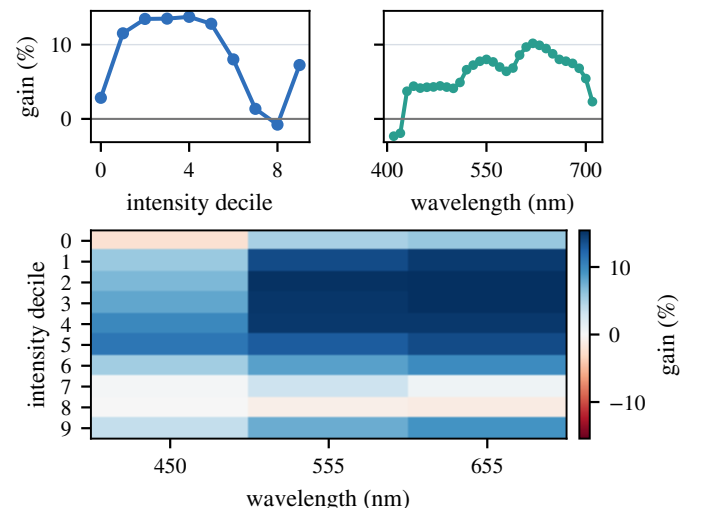


Fig. 6. CAFE error-regime analysis at rank (12, 12, 4). Top: marginal RMSE gains by observed intensity and wavelength. Bottom: joint intensity-spectral bins. Positive values indicate lower RMSE than matched-rank Tucker.

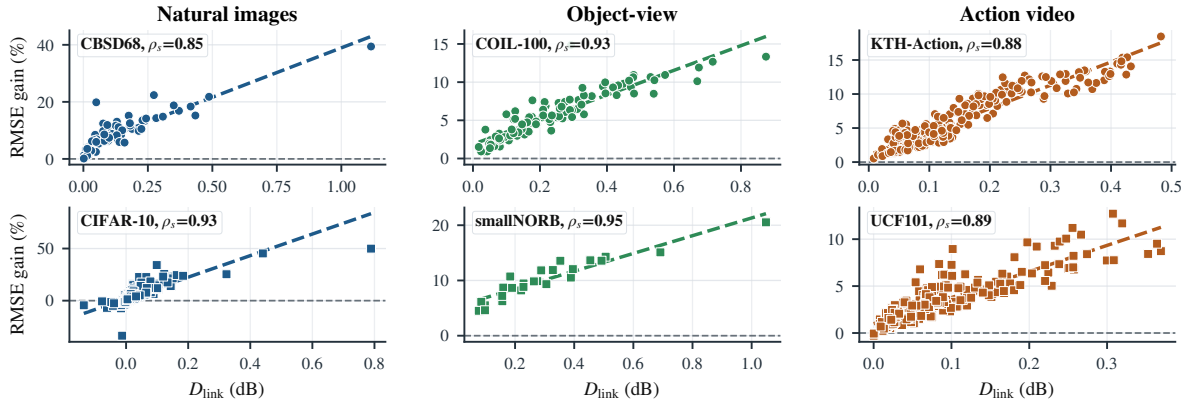


Fig. 7. Cross-domain response diagnostic. Each point plots the diagnostic D_{link} (dB) against the RMSE gain of NTD-PL over Tucker. Larger diagnostic scores predict larger benefits from the learned response.

with gradient-based Tucker updates. The diagnostic measures how much of a fixed Tucker residual can be removed before joint NTD-PL training.

Controlled experiments show that NTD-PL remains close to Tucker for linear measurements and separates as nonlinear energy increases. Hyperspectral reconstruction and completion show improvements in RMSE and spectral angle at matched Tucker rank, while higher-rank Tucker comparisons quantify the extra parameters needed to reach similar RMSE. The mechanism experiments connect the gain to the effective-rank increase after the response, and the cross-domain results show that larger diagnostic scores are followed by larger NTD-PL gains.

Several directions remain open. Monotone splines or physics-informed response families may be preferable when calibration constraints are known. Sharper sample-complexity or identifiability results, richer noise models, and domain-specific response priors are also natural extensions of the framework.

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